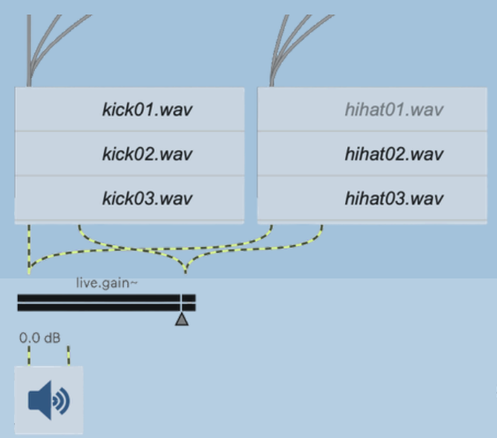
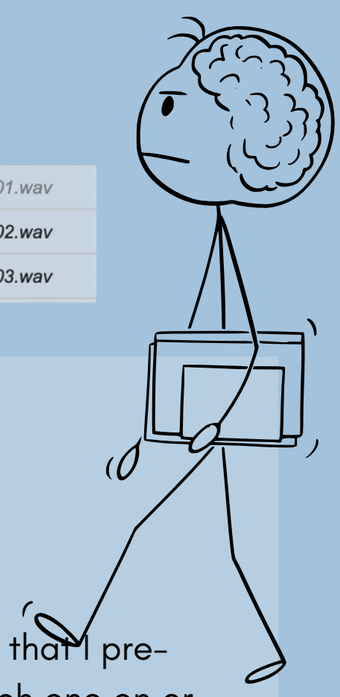
MOOD SWINGER

Amber Shi Summer 2026

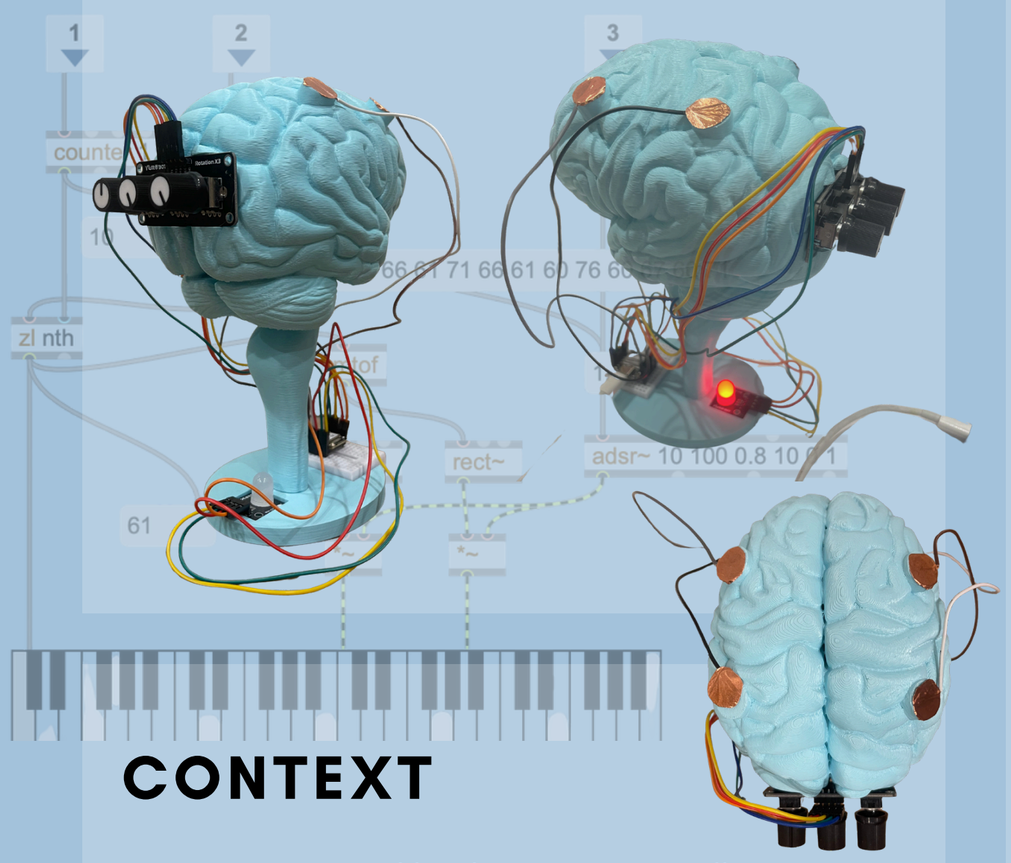




INTRODUCTION

The two hemispheres of the brain listen for different things in music: the left side attends to rhythm while the right side focuses on pitch.

Mood Swinger (see figures below) is a 3d printed brain model controller that allows users to control rhythm and pitch by pressing buttons on the respective hemispheres which detect them. The sounds that these controls play depend on 3 different emotions that the user can select by turning a knob: happy, angry, and confused.



CONTEXT

Existing projects like the Dato DUO allow children/beginning musicians to synthesize and produce their own beats with a partner [1]. Connecting this concept to music psychology, I created a model of the brain so users can create music alone while understanding what the hemispheres of the brain contribute to analyzing music.

Studies show that the brain works in a hierarchy, where sensory input is taken in, then complex pattern perception occurs, followed by associations between multiple sensory areas [2]. The Mood Swinger similarly shows this pattern, as the users can use the model to synthesize among 3 sound patterns depending on the chosen emotions (happy, angry and confused).

IMPLEMENTATION

Interaction System

To play Mood Swinger, users touch sensors on the brain to turn the drums and melodies on and off. Three knobs let them change the mood, the tempo, and the cutoff frequency of a low-pass filter. I wired these controls to a Seeed Studio XIAO ESP32-S3 microcontroller and programmed it in the Arduino IDE. The microcontroller sends the data to Max/MSP through serial communication.

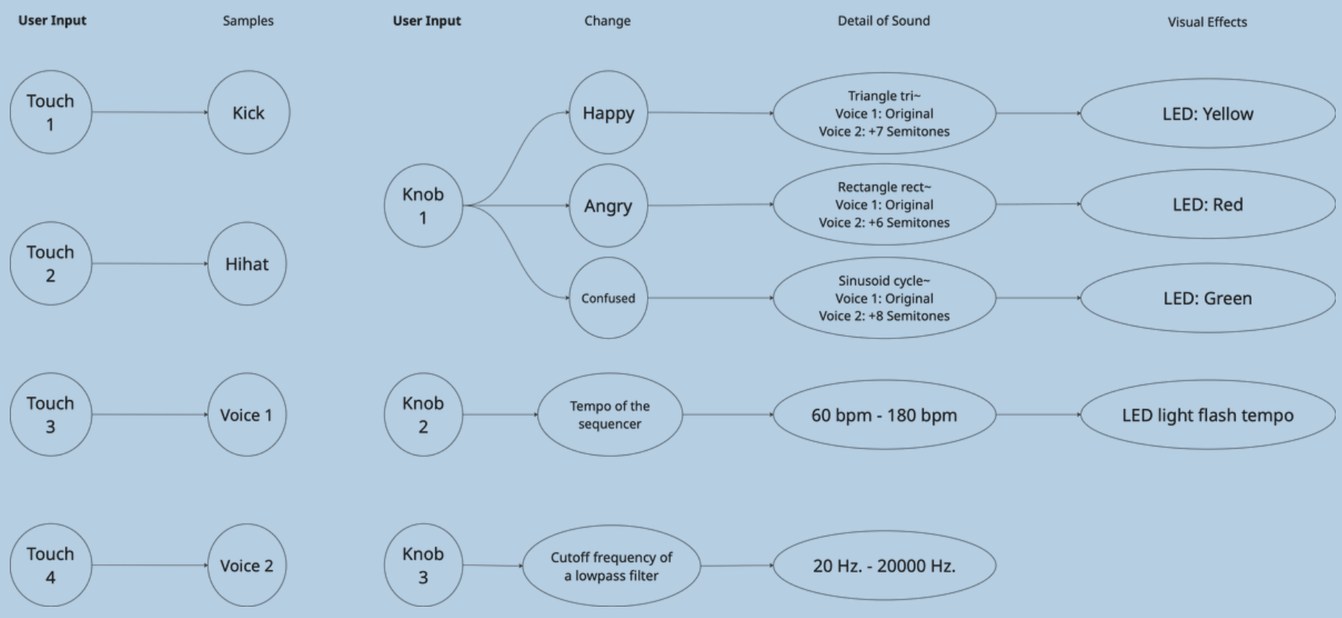
Max/MSP then turns the controls into sound. I added an LED that changes color with each emotion (yellow: happy, red: angry, and green: confused) and blinks along with the tempo of the music.

Audio System

Each emotion has two melodies that I pre-programmed. Users can turn each one on or off with the touch sensors on the right hemisphere.

I gave each emotion its own sound by changing the waveform of the melody. Happy uses a triangle wave, which sounds light and clear. Angry uses a loud, abrupt square wave, and confused uses a quiet, mysterious sine wave. I programmed all of these differences in Max/MSP.

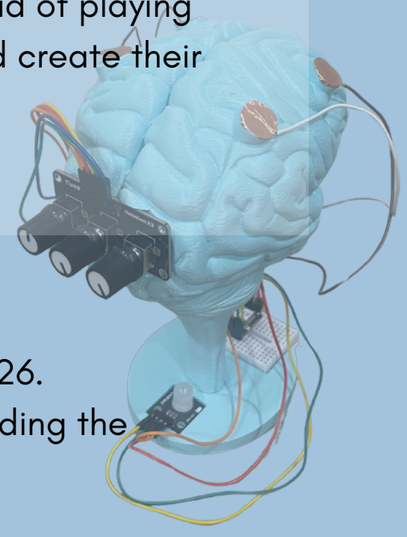
Each emotion also has its own kick and hi-hat, which I took from Ableton Live's sample library. I shaped them with EQ and effects, boosting or cutting certain frequency ranges to match the mood of each emotion. The two touch sensors on the left hemisphere play the drums: one triggers the kick, and the other triggers the hi-hat.



FUTURE WORK

I have a few ideas for improving Mood Swinger. First, to make it easier to use, I want to add small icons of drums and emotions to the touch sensors and knobs, so users can tell right away what each control does.

I also want to build another version that flips the idea around: instead of playing pre-programmed melodies, users would hear a set drum pattern and create their own melody to match it.



REFERENCES

[1] "Dato Musical Instruments." Dato, dato.mu/. Accessed 10 Aug. 2026.
[2] Thompson, William Forde. Music, Thought, and Feeling: Understanding the Psychology of Music. 2016.